

Brain Scans Reveal How Expert Echolocators Build Mental Maps Click by Click

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For some blind individuals, navigating the world doesn't require a cane or a guide dog just a series of sharp tongue clicks. This ability, known as **echolocation**, allows them to sense nearby objects by listening to the returning echoes. Now, scientists have peered inside the brains of expert echolocators to understand how this remarkable skill works in real time.

In a study published in eNeuro, **cognitive neuroscientists** Santani Teng and Haydée García-Lázaro from the Smith-Kettlewell Eye Research Institute recorded brain activity from four blind **echolocation** experts and 21 sighted, inexperienced volunteers. Participants listened to recorded clicks followed by echoes designed to simulate an object to the left or right. After sets of two, five, eight, or eleven click-echo pairs, they indicated the object's direction.

The blind experts vastly outperformed the sighted group. One echolocator could determine the object's location after just two click-echo pairs. **Electroencephalography** (EEG) data revealed why: with each successive click, the experts' brains accumulated **spatial information**, refining their mental map. "Rather than through a single optimal snapshot," explained neuroscientist Monica Gori, who was not involved in the study, "echolocators combine these additional details over time."

This finding challenges the notion that a single echo provides a complete picture. Instead, **echolocation** is a dynamic process of continuous updating. The study also confirmed that **echolocation** activates **visual brain regions**, underscoring the brain's flexibility. Teng emphasized the practical impact: "Echolocators have a truly remarkable skill, with real-life benefits." The team now aims to uncover how experts learn to **suppress** the sound of their own clicks and focus solely on the echoes a crucial step in mastering this extraordinary human ability.

Word Treasure

echolocation -- A technique where animals or people make sounds and listen to echoes to locate objects around them.

electroencephalography -- A method of recording electrical activity in the brain using sensors placed on the scalp (abbreviated EEG).

cognitive neuroscientists -- Scientists who study how the brain enables thinking, learning, and perception.

spatial information -- Data about the position and arrangement of objects in space, like knowing where something is to your left or right.

visual brain regions -- Parts of the brain, usually used for seeing, that can adapt to process other senses like hearing.

suppress -- To consciously block out or ignore a particular sensation, like the sound of your own click.

Background you need

Echolocation is best known from bats and dolphins, but some blind humans have also learned to use mouth clicks or taps to 'see' their surroundings. This skill is rare and takes years of practice, but those who master it can ride bikes, hike, and even play ball sports without sight.

The Smith-Kettlewell Eye Research Institute in San Francisco is a leading center for studying vision and blindness. The study was published in *eNeuro*, a journal that focuses on brain function and learning.

The researchers, Santani Teng and Haydée García-Lázaro, are cognitive neuroscientists who explore how the brain adapts when one sense is lost. They worked with four expert echolocators--all blind--and compared their brain activity to 21 sighted volunteers using EEG, which records brain waves non-invasively.

The brain is remarkably plastic, meaning it can reorganize itself. For example, in blind people, visual areas often get recruited for touch or hearing. This study provides real-time evidence of how that repurposing works when echoes build a mental map.

Understanding how echolocators filter out their own clicks while focusing on the faint echoes could inspire new technologies or training programs to help more blind individuals gain independence.

Break it down

LEAD: For some blind people, tongue clicks replace canes and guide dogs, letting them 'see' with echoes.

+ specific: 'just a series of sharp tongue clicks'

STUDY ANNOUNCEMENT: Scientists from the Smith-Kettlewell Institute used EEG to study expert echolocators in real time.

+ WHO: Santani Teng and Haydée García-Lázaro led the work.

+ PUBLICATION: *eNeuro* journal.

EXPERIMENT: Blind experts and sighted volunteers listened to click-echo pairs and judged object direction.

+ SETUP: Sets of 2, 5, 8, or 11 pairs.

+ PERFORMANCE: Experts vastly outperformed sighted group; one needed only two pairs.

MECHANISM: EEG showed experts' brains accumulated spatial information with each click--like building a map bit by bit.

+ OUTSIDE COMMENT: Monica Gori explained it isn't a single snapshot, but a combining of details over time.

INSIGHT: Echolocation activates visual brain areas, showing brain flexibility.

+ PRACTICAL QUOTE: Teng says the skill has real-life benefits.

OPEN QUESTION: Researchers now aim to uncover how experts suppress the sound of their own clicks to focus on echoes.

Why it matters

This research shows how adaptable the human brain really is. For a teenager, it's a reminder that skills once thought impossible--like navigating without sight--can be mastered with

practice, and that our brains are constantly rewiring themselves based on how we use them.

Test what you remember

1. What sense do expert echolocators rely on to navigate?
 - (A) Smell
 - (B) Touch
 - (C) Hearing echoes from tongue clicks
 - (D) Vibrations through the ground
2. How quickly could one expert determine an object's location?
 - (A) After eleven click-echo pairs
 - (B) After eight click-echo pairs
 - (C) After just two click-echo pairs
 - (D) After only one click
3. What did the EEG data reveal about the experts' brain activity?
 - (A) Their brains got less active with each click.
 - (B) They accumulated spatial information with each successive click.
 - (C) Their visual brain regions shut down.
 - (D) They processed only the first echo they heard.
4. Who is Monica Gori?
 - (A) The lead researcher of the study
 - (B) A blind echolocator
 - (C) A neuroscientist not involved in the study who commented on it
 - (D) A sighted volunteer in the experiment
5. What part of the brain do echolocators use when processing echoes?
 - (A) Motor regions
 - (B) Auditory regions only
 - (C) Visual brain regions
 - (D) Memory centers
6. What future skill do the researchers want to understand better?
 - (A) How echolocators learn to click louder
 - (B) How experts suppress the sound of their own clicks
 - (C) How echolocators learn to use canes
 - (D) How to train dogs to help echolocators

Answer key (try the quiz first!): 1: C 2: C 3: B 4: C 5: C 6: B

Questions to think about

1. Positive: The study validates the extraordinary ability of blind echolocators and provides solid neural evidence. It could lead to better training programs, helping more blind children and adults gain confidence and mobility, reducing reliance on expensive tools or animals.

2. Negative / Critical: Only four expert echolocators were tested, which is a very small sample. Their brains might be unusually plastic, so the findings may not apply to all blind people. Also, focusing on rare experts could distract from more accessible assistive technologies that benefit a wider population.

3. **Neutral / Analytical:** The brain's ability to repurpose visual cortex for hearing is a well-known phenomenon, but this study adds the timeline: it's not an instant picture but a gradual accumulation. That challenges the idea of one perfect 'echo image' and suggests echolocation is more like a streaming process--similar to how vision builds up detail over multiple glances.

4. **Forward-looking:** If scientists can decode exactly how the brain suppresses self-generated clicks while sharpening echo perception, they might build devices that mimic this filtering. That could lead to smart headphones or implants that enhance spatial hearing in noisy environments, benefiting not just the blind but also sighted people in low-visibility conditions.

MY THOUGHTS (at least 20 words):